Crafting a Strong SOP, CV & Finding PhD Supervisors

Lessons from My Journey

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About Me

- *PhD student,* Carnegie Mellon University (2023 -), Language Technologies Institute
- *BSc, Computer Science,* Federal University of Agriculture Abeokuta, Nigeria (2017-2023)
  - Applied to PhD programs twice before being accepted
  - Started my research journey independently while still an undergraduate

About Me

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  - Started my research journey independently while still an undergraduate
- Built a research profile through ML Collective, an open-access research lab
- Successfully navigated the PhD application process despite resource constraints in Nigeria

Today’s Agenda

- Statement of Purpose (SOP)
- Curriculum Vitae (CV)
- Finding and Approaching PhD Supervisors
- Q&A (20-25 minutes)

What is a Statement of Purpose?

- Definition and purpose
- How it differs from other application materials
- Why it matters in PhD applications

SOP: My Journey

- Not a straightforward path – took ~4 years to save for college
- Built some research experience outside traditional academic structures
- Leveraged remote research opportunities and open science communities (ML Collective, Masakhane)
- Developed research skills with limited institutional resources
- Inspirational stuff, but my ass was rejected! :)

SOP: What worked

- ~~Not a straightforward path — took ~4 years to save for college~~
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SOP: What worked

- More research experiences.
- Very specific research interests.
- More technical depth, and cutting away most of the inspirational stuff

SOP: What worked

- **Introduction**: Hook and brief overview
- **Academic/Research Background**: Relevant experiences
- **Research Interests**: What you want to explore and why
- **Program Fit**: Why this specific program is right for you
- **Future Goals**: How the PhD aligns with your career plans
- **Conclusion**: Summarize your fit and readiness

SOP Do’s

- Be specific about research interests
- Connect past experiences to future goals
- Tailor each SOP to the specific program
- Demonstrate knowledge of the department's research
- Show how you'll contribute to the academic community

SOP Don’ts

- Don't include irrelevant personal stories
- Avoid vague, generic statements
- Don't focus too much on undergraduate coursework
- Avoid overly technical jargon
- Don't repeat information from other application materials

SOP Examples: Lessons from My Applications

- Example from my first application (which was rejected)

Statement of Purpose

Growing up in Nigeria, I was always expected to become a civil engineer or a doctor, with no alternative. But during the four years of working after high school to save up for college, I began to read more about artificial intelligence (AI) and its potential applications for societal impacts. As a result, I bought my first computer to teach myself computer science and AI. After learning more about the field, I chose to pursue a Bachelor's degree in computer science at the Federal University of Agriculture Abeokuta (FUNAAB).

During my studies at FUNAAB, I learned that sub-Saharan Africa has the highest population of people with hearing disabilities but the least resources to support sign language training and

SOP Examples: Lessons from My Applications

- Example from my successful application

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I am interested in developing Machine Learning (ML) algorithms that optimize for data, model, and system efficiency; and how these methods solve real-world problems in Natural Language Processing (NLP) and Computer Vision (CV). As Artificial Intelligence becomes increasingly data and computationally driven, the importance of learning data representations with efficiently optimized methods cannot be overemphasized. Also, the rather steep increase in the size and costs of ML models in recent years means that participation in the development of the most prominent models is limited to a small number of teams within resource-rich institutions. The limited resources that have been available to me as an independent researcher have made me very conscious and calculative when designing experiments. During my PhD at CMU, I would like to focus on designing methods that enable us to build efficient models and systems with less computation and data resources, thereby lowering the barrier to shaping the ML technologies we create. Below, I discuss my research experiences and intended future directions.

Data Efficiency: The most significant constraints to developing and deploying real-world ML

SOP Examples: Lessons from My Applications

- How to discuss research without formal experience
- How to connect your unique background to research interests
- How to show passion for solving region-specific problems

CV

- Academic CV vs. Resume
  - Key differences in purpose, length, and content
  - What selection committees look for in a PhD applicant's CV
    - Education
    - Research Experience
    - Publications/Presentations
    - Technical Skills
    - Teaching Experience
    - Awards and Honors

CV: Highlighting Research Experience

- How to describe research projects
- Emphasizing your contributions
- Connecting research to your proposed PhD focus

CV Formatting and Style

- Clean, consistent formatting. **Use Latex!**
- Easy-to-scan structure
- Academic CV conventions
- Length considerations

Common CV Mistakes

- Focusing too much on non-~~academic~~research work
- Poor organization of information
- Lack of specific research details
- Inconsistent formatting
- Undervaluing work on region-specific problems
- Failing to show initiative despite resource constraints

3. Finding Supervisors

- **Finding your advisor matters!**
  - The impact of supervisor selection on your PhD experience
  - The supervisor-student relationship
  - Different supervision styles

How to identify potential advisors

- Using academic search engines/databases
  - For CS folks, using csrankings.org
- Following recent publications in your field
- Conference/workshop proceedings and presentations
- University department pages
- Meet them at conferences

Research Compatibility

- Aligning your research interests
- Publication history and trajectory
- Current projects and funding
- Lab size and composition
- Graduation rate of previous students

Approaching potential advisors

- Crafting effective initial emails
- What to include in your introduction
- How to demonstrate you've researched their work
- Following up appropriately
- Overcoming the "cold email" challenge from Africa
- Leveraging conferences and workshops for connections
- Using LinkedIn and Twitter strategically
- Finding advisors who have worked with African students

Sample email template

Bringing it all together

- Start your preparations early!!!
  - Not by drafting all these materials, but by your research
    - Focus is very important; Experiences are even more important
- Reaching out starts around Fall.

Resources you can use

- Research Experiences & Colabs.
  - CMU!!!
  - Open-access research communities: ML Collective, Masakhane, Cohere for AI, etc
  - Watch out for the Fatima Fellowship program.
  - Africa-specific conferences and workshops: Deep Learning Indaba, AfricaNLP @ICLR; ACL
- SOPs examples: http://cs-sop.org/
- Application support programs: https://github.com/chinasaokolo/csGraduateApps
- General:
  - https://github.com/shaily99/advice
  - Watch out for the annual Black in AI's mentorship program
    - They help with app fee waivers for certain school too

Q&A Slide

Thank you!